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Optimized machine learning framework for cardiovascular disease diagnosis: a novel ethical perspective

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Abstract

Alignment of advanced cutting-edge technologies such as Artificial Intelligence (AI) has emerged as a significant driving force to achieve greater precision and timeliness in identifying cardiovascular diseases (CVDs). However, it is difficult to achieve high accuracy and reliability in CVD diagnostics due to complex clinical data and the selection and modeling process of useful features. Therefore, this paper studies advanced AI-based feature selection techniques and the application of AI technologies in the CVD classification. It uses methodologies such as Chi-square, Info Gain, Forward Selection, and Backward Elimination as an essence of cardiovascular health indicators into a refined eight-feature subset. This study emphasizes ethical considerations, including transparency, interpretability, and bias mitigation. This is achieved by employing unbiased datasets, fair feature selection techniques, and rigorous validation metrics to ensure fairness and trustworthiness in the AI-based diagnostic process. In addition, the integration of various Machine Learning (ML) models, encompassing Random Forest (RF), XGBoost, Decision Trees (DT), and Logistic Regression (LR), facilitates a comprehensive exploration of predictive performance. Among this diverse range of models, XGBoost stands out as the top performer, achieving exceptional scores with a 99% accuracy rate, 100% recall, 99% F1-measure, and 99% precision. Furthermore, we venture into dimensionality reduction, applying Principal Component Analysis (PCA) to the eight-feature subset, effectively refining it to a compact six-attribute feature subset. Once again, XGBoost shines as the model of choice, yielding outstanding results. It achieves accuracy, recall, F1-measure, and precision scores of 98%, 100%, 98%, and 97%, respectively, when applied to the feature subset derived from the combination of Chi-square and Forward Selection methods.

Keywords Artificial intelligence, Cardiovascular diseases, Chi-square, Feature selection, Machine learning, K-nearest neighbours, Principal component analysis, Random forest, XGBoost, Decision trees, Logistic regression

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Introduction

The heart is the most vital organ of the human body. It works like a pump responsible for the circulation of blood in the body of a person. Cardiovascular disease (CVD) is a condition that affects the heart and blood arteries [1–3]. There are several types of cardiovascular or heart diseases; some notable diseases are congestive heart failure, arrhythmias, congenital disease, angina pectoris, myocarditis, cardiomyopathy, heart attack, coronary disease, and heart cancer [4–6]. These diseases occur due to different factors such as an unhealthy diet, excessive use of alcohol or tobacco, physical inactivity, obesity, blood pressure, glucose, and blood lipids [7]. Around 18 million people are killed annually due to this disease [8]. Manual diagnosis takes time and cost to evaluate all the parameters. However, advances in medicine and data availability provide a way for decision support systems. Therefore, to evaluate CVD parameters, automated software can help train and test to help doctors in the decision-making process [8, 9].

The integration of Artificial Intelligence (AI) and Machine Learning (ML) in healthcare has opened new avenues for efficient and timely diagnostics of diseases [10–12]. It is not a question of simply using the technologies for cardiovascular health, but a means to align AI with sociocultural needs in society [13–15]. The availability of medical data creates new opportunities for researchers to align an AI-based system that helps diagnose heart disease patients. Moreover, identifying essential features is also crucial, directly impacting the model's performance. Cutting down the number of features can decrease the complexity of the model and can help in better prediction [16, 17]. With the help of ML, it is possible to identify essential features that can be used to classify these diseases. Numerous lives could be saved by predicting CVD early and efficiently. The prediction of these diseases through ML will help doctors in their decision-making process [18]. Furthermore, time and cost can be minimized because the proposed system will do it automatically instead of manually processing the required tests [19–21].

Despite the advancements in ML applications for CVD classification, existing research often falls short in three key areas: Incorporating ethical AI principles, obtaining model generalizability across several datasets, and making a comprehensive feature selection. However, relying on limited feature selection techniques for many studies can reduce the accuracy, precision and interpretability of the model [22, 23]. Ethics considerations for transparency, fairness and bias mitigation are often ignored in high-stakes medical decision-making. This work crosses these gaps using an integrated feature selection methodology that designs features according to ethical AI principles, producing results robust to many ML models. When developing an ML model, it is seldom the case that all variables in the data set benefit the model. Some features are significant and contribute more than others [19, 24]. Irrelevant features decrease the generalization ability of the model and may reduce the overall accuracy of a classifier [25–28]. Therefore, feature selection have direct impact on model performance [29–31]. This paper identifies significant features through different feature selection techniques and classifies the CVD efficiently. It entails a novel method for using ML techniques to identify and categorize CVD in its early stages. It addresses the urgent need for precise and effective methods to identify individuals at risk of developing CVD to enable prompt interventions and preventive measures. It aims at the ethical AI principles of transparency, interpretability, and bias mitigation in diagnostic models. Therefore, the AI helps to enhance the accuracy and fairness of the models, providing a better basis for practice in healthcare services [32, 33].

The proposed model uses patient data, including demographic details, medical histories, and the findings of diagnostic tests. The relevant and instructive features are derived from the dataset by extensive analysis and feature engineering. Some of these features are then used further by different ML algorithms, such as K-Nearest Neighbours (KNN), Logistic Regression (LR), and Neural Networks (NN). The model is trained and validated on a vast and varied dataset consisting of patients with CVD. The paper gives the performance of the model with several evaluation metrics like accuracy, recall, precision, and F1-score. Results: This model is proven to be useful to classify between normal and diseased participants with high accuracy, giving good information for early diagnosis and individualized treatment. The proposed approach supports the management of cardiovascular health by ensuring identification and the decisions associated with the CVD risks, which consequently improves patient health and a reduction in healthcare costs. This research is inspired by ethical frameworks for AI, such as the EU's Ethical Guidelines for Trustworthy AI and the IEEE's Ethics in AI [34]. To achieve the effectiveness of ethical alignment in CVD diagnostic AI, this study includes fairness-aware algorithms for feature selection while utilizing validation metrics significantly throughout various datasets. Furthermore, techniques of bias elimination are also implemented, for instance, balanced data and interpretability tools to avoid and increase credibility in material developmental applications in the medical informatics field. It ensures that the proposed model aligns with human values and society's needs. This alignment is crucial in medical applications where decisions significantly impact the patient's outcome, and

ethical considerations must guide the adoption of AI technologies.

The following are the contributions of the suggested paper:

1. It aligns an innovative AI-based model with early classification and detection of CVDs.
2. The proposed model employs a combination of DT, RF, XGBoost, KNN, LR, and NN algorithms for analysis and categorization of patient data. The steps on how to remove irrelevant and non-informative features in the dataset are well-articulated to assure that the model gets the precise inputs to provide a precise classification and prediction of the heart condition.
3. It underlines the vital importance of the feature selection and reasons for the chosen features. For the proposed model, a rigorous test and validation is executed with a large and diverse dataset of patients suffering from CVDs.
4. It demonstrates model performance and efficacy to accurately classify patients on offer of explicit evaluation metrics, such as accuracy, precision, recall, and F1-score.
5. It addresses complexity and heterogeneity in the data of CVDs through employment of different ML algorithms.
6. It provides insights into the pros and cons of each strategy through the contrast and evaluation of the performance of different algorithms. The model presented further carries practical implications in treating CVD at the early stage. The model can thus help in the identification of those at risk, thereby prompting interventions and generation of personalized treatment plans by the appropriate classification of subjects into healthy and diseased.

The rest of the article is structured into different sections; The related work is analyzed and presented in "Related work" section. The proposed methodology is provided in "Proposed methodology" section. The results and discussion are provided in "Experimentation setup, results, and discussion" section. Finally, the article is concluded in "Conclusion" section.

Related work

The integration of Machine Learning and other AI techniques in the healthcare are extensively studied for this research. Some of the most relevant literature are discussed in this section.

An ML-based model for heart disease classification using Principal Component Analysis (PCA) for dimension reduction and Chi-Square for feature selection is proposed by Garate-Escamila et al. in [8]. The authors performed four experiments using three datasets (Cleveland, Hungarian, C-H). Firstly, ML classifiers were applied directly without any feature selection or reduction technique. The second experiment uses the chi-square technique to obtain essential features. The features extracted through chi-square were reduced by applying PCA. In the final experiment, PCA was applied to raw data directly. After preprocessing, multiple ML models, such as LR, Naive Bayes (NB), DT, RF, Gradient-Boosted Tree (GBT), and Multilayer Perceptron (MLP), were applied. Random Forest (RF) reported the highest accuracy score of 98% among others; however, the small size of the dataset has been used. However, it must be tested with a large dataset for better generalization. Amin et al. focused on identifying significant features for predicting heart disease using a brute force technique [16]. Different combinations of a minimum of three features were made, and then different ML models (DT, KNN, LR, SVM, Vote, and Neural Nets) were observed in combination with each set of features. A hybrid model (Vote) outperformed other models by scoring a superior precision of 87% on nine features. The brute force technique is not feasible and can only work with a few features. The authors in [9] proposed a method for optimal feature selection using a hybrid approach called Cluster-based DT Learning (CDTL) for CVD classification. The authors minimise feature dimensionality by 61.53% without compromising prediction precision. In [35], the authors proposed an optimized technique for feature selection using SVM and Genetic Approach (GA) to predict Heart diseases (HDs) efficiently with minimum features. The authors used different techniques for feature selection, such as Chi-Squared (CS), Information Gain (IG), Filtered Subset (FS), One-Attribute Based, Relief, Gain Ratio (GR), Filtered Attribute, Genetic Algorithms (GA), and Consistency Subset, Correlation-Based Features Selection (CFS). The performance of different classifiers like SVM, KNN, MLP, and J48 are compared by giving the original dataset in which SVM outperformed others by scoring an accuracy of 83.70%. Afterwards, these models are given significant features obtained through each method, and their performances are compared. The combination of SVM and GA outperformed other techniques by reporting the highest accuracy of 88.34%.

Multiple techniques such as PCA, Correlation-based FS (CFS), Cluster-based FS (LLCFS), and Infinite Latent Feature Selection (IFLS) are utilized for feature selection [36]. The authors also used different variants of SVM (Linear, Gaussian, Polynomial, and Sigmoid) along with Naïve Bayes and LR for HDs prediction. Two experiments were conducted using three different HDs datasets. In the first experiment, all the classifiers were trained on the data. Then their performances were compared with

each other, in which the Linear SVM performed well by reporting an accuracy, specificity, sensitivity, and AUC of 89.93, 0.93, 0.87, and 0.96, respectively. Multiple techniques for feature selection were used in the second experiment, in which the ILFS gave better results by reducing 58 features to 39 without compromising model performance. Using 39 features for CVD classification is feasible; however, there is still room for feature reduction to a smaller number with stable results. [37] used three feature selection techniques: Lasso Regularization, Pearson Correlation (PC), and Recursive Features Elimination (RFE) with different ML algorithms for prediction [38]. The combination of Pearson correlation and lasso regularization with RF performed well compared to others, giving an accuracy of 99%. Although the model gives better results, it tends to overfit the new examples. Performed a comparative analysis of various ML algorithms, such as LR, DT, NA, ANN, and a novel K-means Neighbors (K-MN) technique to predict heart disease. K-MN generated the highest accuracy, which was around 93%. However, the efficiency of the suggested method needs to be evaluated along recall and precision. Sharma and Parmar used a deep NN on the Cleveland dataset and compared it with other ML classifiers [39]. They optimized the hyperparameters of the DNN using Talos, resulting in an accuracy of 90.78%. In [40], six ML models were used for CVD prediction, including LR, KNN, SVM, ANN, NB, and RF. Various data mining tools were also utilized in the experiment, such as Weka, RapidMiner, Knime, Orange, Scikit-Learn, and Matlab. The accuracy of each classifier was compared on each tool, and the highest accuracy of 85.86% was achieved by ANN using Matlab as a data mining tool. Idris et al. [41] employed RFE, Chi-square, and embedded DT methods for feature selection while using SMOTE to balance the dataset. After preprocessing, KNN, LR, NN, DT, SVM, Vote, and DL were applied to the extracted features. The NN performed the best overall on both datasets, particularly on the extracted features through embedded DT compared to other models, achieving an accuracy of 94.5%, precision of 94.5%, recall of 95.1%, and F1-measure of 94.5% on the UMMC dataset, and accuracy, precision, recall, and F1-measure of 89.7%, 89.7%, 91.6%, and 89.7%, respectively, on the SAH dataset.

The above-discussed studies show improvements in applying ML and feature selection techniques for CVD classification, as summarized in Table 1. Methods such as PCA, Chi-square, Genetic Algorithms, and Recursive Feature Elimination help reduce dimensionality and enhance the model's performance. RF and XGBoost ensemble models, as well as DNNs, have yielded high accuracy in heart disease prediction. Several limitations remain, however. The generalizability of many studies is affected due to small or imbalanced datasets. Typically, feature selection techniques focus on computational efficiency but disregard whether the procedure followed is ethical, such as whether it considers transparency or whether it can mitigate bias. Additionally, the tradeoffs between model complexity and interpretability have not been well explored, especially for real-world applications in resource-constrained spaces. To address the above gaps, this thesis integrates advanced feature selection methods with ensemble models, applies fairness-aware mechanisms, and validates the approach on more extensive and diverse datasets to provide robust and trustworthy predictions.

Proposed methodology

The flowchart of the proposed work for classifying CVDs is illustrated in Fig. 1. Initially, the Heart Disease dataset is obtained from the UCI repository [45] and then preprocessed by applying various filter and wrapper techniques to extract the relevant features that contribute more to the output class. After applying PCA to reduce data dimensions, the extracted features are fed into an ML model. This study used hyperparameter optimization to improve the performance of the developed ML models. This approach employed a systematic grid search strategy with detailed tuning of critical parameters for each model. For instance, parameters such as *learning_rate*, *max_depth*, and *n_estimators* in XGBoost were fine-tuned, while for RF, *n_estimators*, *max_features*, and *min_samples_split* were optimized. Each combination of hyperparameters was tested using 5-fold cross-validation on the training dataset. The F1 score was chosen as the primary evaluation metric since it balances precision and recall. The final optimized hyperparameters are shown in the evaluation section, along with the model results. The datasets used in the proposed work are accessed from an ML repository called UCI [46]. The datasets used in this study originate from the UCI Machine Learning Repository and include records from four databases: Hungary, Cleveland, Switzerland and the VA Long Beach. The most used dataset is Cleveland, which has 303 samples with 14 main attributes used in ML experiments. Several demographic features, clinical physiological characteristics, and diagnostic information, including cholesterol level, maximal HR, resting blood pressure and rest thalassemia status (thal), as well as maximal heart rate, resting blood pressure, and rest thalassemia status (thal), are considered these attributes.

The datasets represent multivariate, heterogeneous patient populations ranging in age from 29 to 77, with a male-to-female ratio of approximately 2:1. The Cleveland dataset is completed; however, missing values in other datasets suggest the need for preprocessing techniques

Table 1 Summary of recent Literature w.r.t Dataset, used methods and limitations

Ref.	Dataset	ML classifier	Feature selection methods	Method	Evaluation parameters	Limitations
[8]	Cleveland Hungarian	DT, RF, LR, NB, GBT, MLP	PCA, Chi-square	RF + Chi-square with PCA	Accuracy = 98% Precision = 100% Recall = 97%	Chi-square applied on continuous features.
[16]	Cleveland	KNN, DT, LR, NB, Vote, SVM, NNs	Brute Force	Vote + Brute Force	Accuracy = 78% Precision = N/A Recall = N/A	The proposed method for feature selection is not feasible for high-dimensional datasets.
[9]	Cleveland	HRFLM, DT RF, SVM, NA, NN, KNN	DT	HRFLM + DT	Accuracy = 88% Precision = 82% Recall = 90%	Need to use other feature selection techniques.
[35]	Cleveland	SVM, MLP KNN, J48	Chi-square, Info gain, Ratio gain, CFS, GA	SVM + GA	Accuracy = 88% Precision = N/A Recall = N/A	Model needs to be trained on the large dataset for better generalization.
[36]	Cleveland Hungarian Switzerland	SVM, NB, LR	PCA, CFS, LLCFS, ILFS	Linear SVM + ILFS	Accuracy = 90% Precision = 90% Recall = 91%	Model generalizability.
[38]	UCI	Tree based Algorithms	Pearson Correlation, RFE, Lasso	RF + Pearson Correlation	Accuracy = 90% Precision = 90% Recall = 99%	Prone to overfitting.
	Cleveland	DT, NA, ANN, LR, K-MN	Lasso Relief MRM	KMN	Accuracy = 93% Precision = N/A Recall = N/A	Less number of evaluation parameters.
[39]	Cleveland	LR, RF, SVM, NB, KNN	None	DNN	Accuracy = 90.78% Precision = N/A Recall = N/A	Need to evaluate the model with other parameters.
[40]	Cleveland	LR, KNN, SVM, ANN, RF, NB	None	ANN	Accuracy = 85.86% Precision= 87.50% Recall = 83.94%	Complex model has been used.
[41]	UMMC SAH	KNN, LR, NN, DT, SVM, Vote and DL	RFE, Chi-square Embedded DT	NN + Embedded DT	Accuracy = 94.5% Precision= 94.5% Recall = 95.1%	Features have been extracted through DT, while for classification, NN is used. This increase model complexity in term of time and resources.
[42]	Cleveland	KNN, CNN, LR, NB, SVM	None	CNN	Accuracy = 94%	Not suitable for tabular data
[43]	Cleveland	KNN, NB, DT, RF	None	KNN	Accuracy = 90% Precision = N/A Recall = N/A	Feature selection technique can be used for better results.
[44]	Cleveland	LR, SVM, ANN, NB, DT, RF	None	ANN	Accuracy = 89% Precision = 98% Recall = 76%	Time complexity.

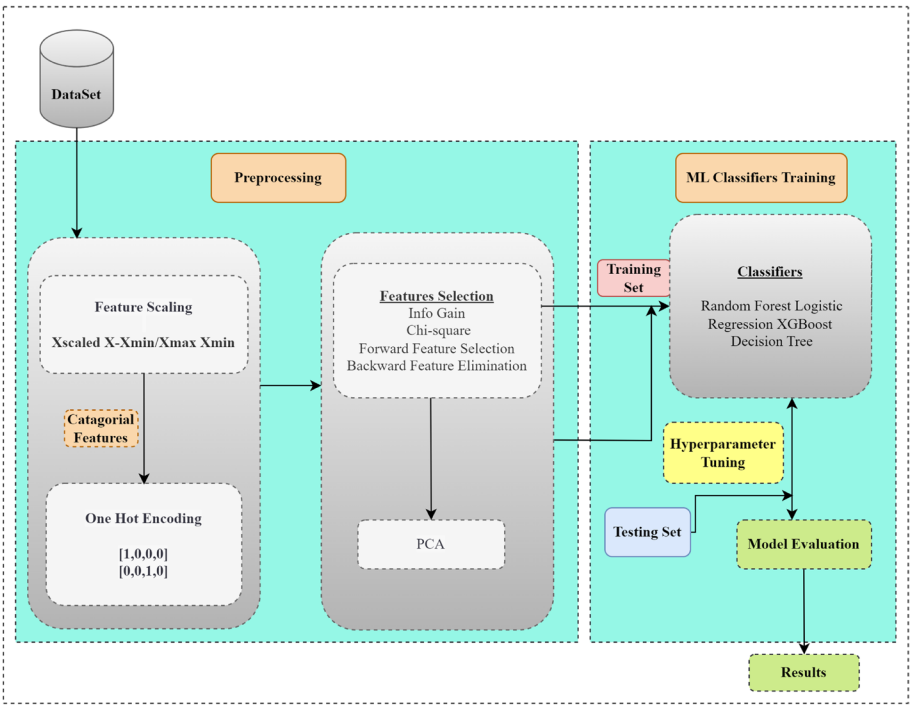


Fig. 1 Conceptual model depicting the workflow for CVD classification, including data preprocessing, advanced feature selection (e.g., Chi-square, Info Gain), dimensionality reduction (PCA), and model evaluation with ethical AI considerations

that help prepare data for robust and reliable analysis. For this study, the combined dataset was split into an 80% training set and a 20% testing set to ensure the proposed methodologies were thoroughly evaluated. The repository contains dataset of different heart disease, such as Long Beach, Hungarian, Cleveland and Switzerland. These datasets contain features such as fbs, age, cp, sex, thal, chol, exang, thalach, oldpeaktrestbps, ca, restecg, slope, and target. The details are summarized in Table 2. Table 3 provides the symbols and their description later used in this section. The proposed scheme can be concluded in the following main steps: (1) data preprocessing; (2) feature selection; (3) significant feature; (4) model building; (5) ML classifiers.

Data preprocessing

For the ML model to function correctly, data must be clean and in the correct format. If we give the data to an ML model without performing the required preprocessing steps, then the model will not work well [47, 48]. Therefore, before implementing any ML project, preprocessing must be done [49]. The dataset has a near-balanced class distribution that eliminated the need for additional class balancing techniques, such as SMOTE or undersampling. Nevertheless, more advanced preprocessing measures were applied, such as outlier detection and removal (e.g., for cholesterol and resting heart rate), to increase model robustness. Additionally, feature normalization is applied for highly variable features, such as cholesterol and resting blood pressure. These measures guaranteed the same model performance for all features. The following substeps are performed in data preprocessing:

Table 2 Description of the dataset used for the proposed methodology

Feature name	Type	Feature description
age	Numeric	Age of the patient
sex	Categorical	Gender (Male or Female)
cp	Categorical	Pain type (1, 2, 3 or 4)
trestbps	Numeric	Resting blood pressure value of an individual
chol	Numeric	Serum cholesterol in mg/dl
fbs	Categorical	Value of an individual with 120mg/dl
restecg	Categorical	Resting electrocardiographic results
thalach	Numeric	Max heart rate achieved by an individual
exang	Categorical	Angina induced by exercise (0 or 1)
oldpeak	Numeric	Numeric value measured in depression
Slope	Categorical	The slope of the peak exercise
ca	Categorical	majorvessels
thal	Categorical	thalassemia
target	Categorical	thalassemia

Table 3 Nomenclature

Symbol	Definition
D	Dataset
$\emptyset$	Missing value or special character
$\hat{D}$	Cleaned dataset
C	Categorical variable
$\hat{C}$	Encoded categorical variable
F_i	i-th feature in the dataset
$\hat{F}_i$	Scaled feature
D_0	Instances of the majority class
D_1	Instances of the minority class
X_i	i-th input feature
Y	Target variable
$Score(X_i, Y)$	Statistical measure for feature X_i and target Y
S	Subset of features
$Performance(S)$	Performance measure of a model with feature subset S
X	Feature matrix
$\hat{X}$	Transformed feature matrix using PCA

Data cleaning

Data cleaning attempts to remove or modify instances of missing values, including null values, spaces, or any other special characters. Data needs to be clean and must be in an ML model-compatible format. If we give the data to any ML model without performing the required preprocessing steps, the model will not work well. Therefore, before the implementation of any ML project, data must be cleaned, and null values should be removed or replaced [50]. The dataset used in this experiment is clean, having no null values. Let D be the dataset. Data cleaning involves handling missing values, null values, spaces, or special characters within D. The cleaned dataset is denoted as $\hat{D}$.

$$\hat{D} = D_{cl}(D) \tag{1}$$

The function $D_{cl}(\cdot)$, in Eq. 1, represents the data cleaning process applied to the dataset D, resulting in the cleaned dataset $\hat{D}$.

Handling categorical variables

Label data rather than numeric values are referred to as categorical data. Categorical variables, such as nominal variables, are what they are called. Categorical data may be handled directly by specific algorithms [51]. A DT, for example, may be trained directly from categorical data without requiring data transformation. Label data can only be used indirectly by some ML techniques. They demand that all input and output variables be numeric. Such variables can be transformed through one-hot or label encoding [52]. In the proposed work, we have used a one-hot encoding technique that changes the categorical variable by adding a binary value for each integer value. In the given dataset, features such as sex, cp, exang, fbs, restecg, and slope are categorical. All these variables are handled through one-hot encoding. Let C be the set of categorical variables in $\hat{D}$. Categorical data are variables that have label values rather than numeric values. To handle categorical variables, one-hot encoding is applied, which transforms each categorical variable into binary values. Let $\hat{D}_{encoded}$ represent the dataset after applying one-hot encoding to C.

$$\hat{D}_{encoded} = OneHotEncoding(\hat{D}, C) \tag{2}$$

The function $OneHotEncoding(\cdot)$, in Eq. 2, represents the one-hot encoding process applied to the cleaned dataset $\hat{D}$, with respect to the categorical variables C, resulting in the encoded dataset $\hat{D}_{encoded}$.

Feature scaling

Scaling the features in a dataset such that they all fall inside a single range is known as feature scaling. It has dealt with significant changes in magnitude, value, or unit during data preprocessing. If feature scaling is not done, an ML algorithm will presume bigger values to be greater and smaller values to be lower, regardless of the unit of measurement [53]. Most people resort to some data normalization or standardization to scale features. We utilize normalization to confine our data to a range between two values, such as (0, 1) or (−1, 1), whenever required. Standardizing our data gives us a 0 mean with 1 for the variance [54]. In this given dataset, a few features, such as age, resting bps, cholesterol, and max heart rate, have different ranges. We have used a normalization technique, such as a min-max scalar, for scaling the above features. This method rescales a feature value using a distribution value of 0 to 1.

Let F be the set of features in $\hat{D}encoded$. Feature scaling is performed to bring all features in F within a similar range. It is achieved through normalization, specifically min-max scaling. The normalized dataset is denoted as $\hat{D}scaled$.

$$\hat{D}scaled = MinMaxScaling(\hat{D}encoded, F) \tag{3}$$

The function $MinMaxScaling(\cdot)$, in Eq. 3, represents the min-max scaling process applied to the encoded dataset $\hat{D}encoded$, with respect to the features F, resulting in the scaled dataset $\hat{D}scaled$.

Normalization

The dataset used in this study exhibited a near-balanced class distribution, with a 49:51 ratio of cases (i.e., class 0)

to controls (i.e., class 1) for CVD. To maintain this balance across the entire experimental pipeline, stratified sampling was used when splitting the data into training and testing sets. Stratification ensured that both sets maintained the same class distribution as the original data, so class imbalance was not incurred during data division. It was achieved because the dataset was divided into two classes. The balanced nature of the classes meant that features such as Synthetic Minority Oversampling (SMOTE) and undersampling were not necessary. This internal balance, along with the stratified split, minimized the probabilities of class bias toward the majority while retaining the ability to detect the minority class.

Normalization also makes sure that features with different ranges, for example, cholesterol and rest blood pressure are brought into a similar scale. This is especially useful in models which are affected by scaling features such as Logistics regression and neural network by avoiding their inclination towards larger scaled features hence making them more stable and unitive during training and evaluation. Stratified sampling was used to preserve the original class distribution during the training and testing splits. This made it possible that both of the sets contained samples that were quite general of the whole dataset to minimize on biases when evaluating model performance. Despite having clean and balanced data in the given dataset, more strategies like eliminating the outliner point for numerical feature (cholesterol level and resting heart rate) and more advanced form of imputation for any missing value in other datasets would expands the data quality and model stability.

Feature selection

It involves removing non-significant features to decrease the quantity of input features. The process of feature selection contributes to the mitigation of overfitting, enhancement of model performance, and reduction of computational expense [55]. If we use the hit-and-trial approach for feature selection, testing all possible feature combinations will take too much time and processing power. Therefore, feature selection techniques help us find the minor feature set quickly. For supervised problems, the feature selection techniques are classified into three categories: wrapper, filter, and intrinsic methods [49]. In filter methods (Chi-square and Info gain), statistical techniques are used to find and evaluate relationships b/w input and target features. Based on these scores, the critical features are identified. Wrapper methods (forward and backward) use different subsets of the input features and then choose the subset which performs better with the learning algorithm [56]. Some ML models automatically perform feature selection while learning the data.

This approach refers to the intrinsic or embedded method. In this research, filter and wrapper methods have been used for feature selection. Each method is applied to the dataset, and their correlation with the target variable is calculated. Features having the highest correlation with the dependent variable are considered for model training. Among all techniques, info-gain performed well in identifying the significant features. Feature selection aims to identify the most informative features from the dataset that contribute significantly to the classification of CVDs. Various feature selection techniques can be employed, such as chi-square, recursive feature elimination (RFE), or forward feature selection. Let M represent the feature selection method.

$$S_{selected} = FeatureSelection(\hat{D}_{balanced}, M) \tag{4}$$

The function $FeatureSelection(\cdot)$, in Eq. 4, represents the feature selection process applied to the balanced dataset $\hat{D}balanced$ using the method M, resulting in the selected features $S\ selected$. The significant features identified from the feature selection step are combined into a final set of features, denoted as S_{final}. This set represents the most relevant features for the classification of CVDs.

$$S_{final} = CombineFeatures(S_{selected}) \tag{5}$$

The function $CombineFeatures(\cdot)$, in Eq. 5, combines the selected features $S_{selected}$ into the final set of features S_{final}.

Feature selection methods were selected depending on their performance on a mixed dataset containing categorical and numerical variables. Chi-square and Info Gain are more helpful in finding the best categorical features that separate the data well. At the same time, Forward Selection and Backward Elimination help select the best subset using performance measurement at every stage. All these methods were considered suitable for the given dataset since it has balanced classes and varying types of features that the methods need to analyze to balance between model interpretability and efficiency.

Principal component analysis

After selecting the best subset features, we can reduce the number of input features by applying a dimension-reduction technique. PCA is a method that is used to reduce dimensions in a dataset. It transforms a large number of features in a dataset into a small number having most of the information in the data preserved [57]. Dimension reduction results in the loss of some information which can affect model performance. However, PCA works pretty well by doing a little trade-in accuracy for simplicity. PCA follows a few steps to move from high-dimensional data to lower

dimensions [58]. The first step of PCA is data normalization which brings all the data into scale similar for each feature. After data scaling, the computation of the covariance matrix is done. It helps to identify correlated features. High-correlated features often contain biased and redundant information. To determine principle components, eigenvalues, and eigenvectors are computed from the covariance matrix. Principle components are new variables that are constructed by combining initial features. All the new features contain squeezed and compressed useful information from initial features. In the end, the eigenvector having a high eigenvalue is considered the first principle component containing maximum helpful information. All the components are sorted in descending order based on their eigenvalue. The greater eigenvalue, the better the principle component is. Data can be mathematically standardized as follows:

$$S = \frac{A - \overline{A}}{\sigma} \tag{6}$$

where A is the variable value, $\overline{A}$ is the mean, and standard deviation as σ(see Eq. 6). The covariance matrix can be mathematically computed as follows:

$$COV = \frac{\sum(X - \overline{x})(Y - \overline{y})}{n - 1} \tag{7}$$

where $\overline{x}$ and $\overline{y}$, in Eq. 7, denote the mean of X and Y with n number of data points. PCA was conducted on the eight feature subsets to reduce the redundant features and maintain the features with the most variance within the dataset. They do this by balancing the trade-off of making a model simpler and losing information. We focused on the top six principal components and ensured that over 95% of the variance was preserved for the model to retain its diagnostic power. PCA inherently abstracts original features into combined components. However, this transformation had a rational basis, which increased interpretability and reduced follow-on processing computation. It is used to overcome potential redundancies and multicollinearity among features. It preserves more than 95% of the variance and retains the most information while reducing the computational overhead. This is particularly valuable in resource-constrained environments where simplicity and efficiency are critical.

Similarly, applying forward selection and backward elimination brings more accuracy because both methods involve testing features, which contributes significantly to classification models. For the CVD dataset, these techniques are effective due to having relatively mixed categories of possible inputs. These include categorical inputs such as gender, chest pain type, and other potentially continuous but not necessarily uniformly critical inputs like cholesterol levels or resting blood pressure. Cutting and pasting features to enhance accuracy helps eliminate noise and overfitting. It makes models function with fewer interpretational problems of particular relevance for CVD classification; selecting meaningful features that remain after feature deletion boosts diagnostic accuracy and computational speed.

Significant features

This research uses wrapper and filter feature selection methods, such as chi-square, info gain, backward elimination, and forward selection, to identify important features. Each method obtained a subset of 8 more significant features from the entire dataset. A list of the best-selected features by each feature technique is given below in Table 4.

Model building

The next and most critical stage, following data preprocessing, is to construct an acceptability prediction model utilizing various ML algorithms. A multitude of algorithms are available for the construction of models; however, the accuracy of predictions differs among algorithms. Therefore, it is critical to choose the appropriate algorithm, taking into account the nature of the data and the intended recipient. The final dataset currently comprises 1025 instances, each of which has eight attributes. The proposed methodology involved the application of multiple ML classifiers to the distinct sets generated via feature selection techniques. The final set of features S_{final} is used to build ML classifiers for the classification of CVDs, as shown in Eq. 8. Various ML algorithms can be employed, such as LR, support vector machines, orRFs. Let M_{final} represent the ML classifier trained using S_{final}.

Table 4 List of the features selected by each method

Feature selection method	Feature set
Chi-square	age, cp, fbs, thalach, exang, slope, chol, trestbps
Info gain	age, sex, thalach, exang, old peak, slope, cp, chol
Forward Selection	age, sex, restecg, thalach, cp, trestbps, old peak, slope
Backward Elimination	age, trestbps, old peak, restecg, sex, cp, exang, slope

$$M_{final} = ModelBuilding(\hat{D}balanced, Sfinal) \quad (8)$$

The function $ModelBuilding(\cdot)$ trains the ML classifier using the balanced dataset $\hat{D}balanced$ and the final set of features $S\ final$, resulting in the trained classifier M_{final}.

The trained classifier can then predict the presence or absence of CVDs for new, unseen instances, as shown in Eq. 9.

$$\hat{Y} = ModelPrediction(M_{final}, X_{unseen}) \quad (9)$$

The function $ModelPrediction(\cdot)$ predicts the CVD presence or absence for new, unseen instances X_{unseen} using the trained classifier M_{final}, resulting in the predicted labels $\hat{Y}$.

The overall functionality of the proposed model is depicted in Algorithm 1, whereas details of the symbols are given in Table 5. Using various ML models, it aims to classify heart disease based on a given dataset $\mathcal{D}$. The algorithm consists of four main steps. The first step is the feature selection step. In this step, feature selection is performed on the training dataset $\mathcal{D}_{tr}$ to identify the most relevant features for heart disease classification. The selected features are stored in $\mathcal{F}$. In the second step, the preprocessing is done on the training dataset $\mathcal{D}tr$ and the test dataset $\mathcal{D}tt$ using the previously selected features $\mathcal{F}$. The preprocessed training dataset is denoted as $\mathcal{D}tr^{pr}$, and the preprocessed test dataset is denoted as $\mathcal{D}tt^{pr}$. In the third step, different models of training and evaluation are conducted. Each model is trained on $\mathcal{D}tr^{pr}$, and then evaluated using the test dataset $\mathcal{D}tt^{pr}$. The evaluation metrics for each model include accuracy (ac), F1-measure (fm), recall (rc), and precision (pr). Finally, in the fourth step, the best model is selected based on the evaluation metrics. It iterates through each model and each evaluation metric and compares the metric with the current best metric. The best model and metric are updated if a higher metric is found. It provides a systematic approach to classifying heart disease by selecting relevant features, preprocessing the data, training multiple models, and selecting the best model based on evaluation metrics. This approach can help accurately diagnose and predict heart disease based on the given dataset.

Table 5 Symbols and descriptions

Symbol	Description
$\mathcal{D}$	Input dataset given to the ML-based model for training
r	Train-test split ratio
T	Number of iterations
$\mathcal{D}_{tr}$	Training dataset
$\mathcal{D}_{tt}$	Testing dataset
$\mathcal{F}$	Set of selected features
$\mathcal{D}_{tr}^{pr}$	Preprocessed training dataset
$\mathcal{D}_{tt}^{pr}$	Preprocessed testing dataset
DT	DT model
RF	RF model
XGB	XGBoost model
KNN	KNN model
LR	LR model
NN	NN model
ac_{DT}	Accuracy of the DT model
fm_{DT}	F-measure of the DT model
rc_{DT}	Recall of the DT model
pr_{DT}	Precision of the DT model
ac_{RF}	Accuracy of the RF model
fm_{RF}	F-measure of the RF model
rc_{RF}	Recall of the RF model
pr_{RF}	Precision of the RF model
ac_{XGB}	Accuracy of the XGBoost model
fm_{XGB}	F-measure of the XGBoost model
rc_{XGB}	Recall of the XGBoost model
pr_{XGB}	Precision of the XGBoost model
ac_{KNN}	Accuracy of the KNN model
fm_{KNN}	F-measure of the KNN model
rc_{KNN}	Recall of the KNN model
pr_{KNN}	Precision of the KNN model
ac_{LR}	Accuracy of the LR model
fm_{LR}	F-measure of the LR model
rc_{LR}	Recall of the LR model
pr_{LR}	Precision of the LR model
ac_{NN}	Accuracy of the NN model
fm_{NN}	F-measure of the NN model
rc_{NN}	Recall of the NN model
pr_{NN}	Precision of the NN model

Algorithm 1 Heart disease classification algorithm

Input: Dataset $\mathcal{D}$, train-test split ratio r, number of iterations T
Output: Best-performing model and its evaluation metrics
Step 1: Train-Test Split
Split the dataset: $\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{tt}}$ = trainTestSplit($\mathcal{D}, r$)
Step 2: Feature Selection
Select features: $\mathcal{F}$ = featureSelection($\mathcal{D}_{\text{tr}}$)
Step 3: Data Preprocessing
Preprocess data:
$\mathcal{D}_{\text{tr}}^{\text{pr}}$ = preprocess($\mathcal{D}_{\text{tr}}, \mathcal{F}$)
$\mathcal{D}_{\text{tt}}^{\text{pr}}$ = preprocess($\mathcal{D}_{\text{tt}}, \mathcal{F}$)
Step 4: Model Training and Evaluation
for each model M $\in$ {DT, RF, XGB, KNN, LR, NN} **do**
  Train the model: M = trainModel($\mathcal{D}_{\text{tr}}^{\text{pr}}, T$)
  Evaluate the model:
  $\text{ac}_{\text{M}}, \text{fm}_{\text{M}}, \text{rc}_{\text{M}}, \text{pr}_{\text{M}}$ = evaluate(M, $\mathcal{D}_{\text{tt}}^{\text{pr}}$)
end for
Step 5: Model Selection
Initialize: $best_{Model} \leftarrow \varnothing$, $best_{Metric} \leftarrow 0$
for each model M **do**
  for each metric $m \in$ {ac, fm, rc, pr} **do**
    if $m_{\text{M}} > best_{Metric}$ **then**
      $best_{Model} \leftarrow$ M
      $best_{Metric} \leftarrow m_{\text{M}}$
    end if
  end for
end for
Output: $best_{Model}$ and $best_{Metric}$

Train test split

In the initial phase of model development, the dataset is partitioned into training and testing sets. The train-test split, which divides the dataset into training and testing subsets, is a crucial stage in the model-building phase. Let D represent the dataset. The dataset is split into the training set (i.e., D_{tr}), and the testing set (i.e., D_{tt}). As discussed earlier, 20% of data from the dataset is held out for model evaluation. The rest of 80% of the data is for model training. All the ML classifiers are trained on 80% and are evaluated on 20% of the data.

Modeling

In the initial phase of model development, the dataset is partitioned into training and testing sets. The train-test split, which divides the dataset into training and testing subsets, is a crucial stage in the model-building phase. After training, different parameters, such as accuracy, F1-measure, recall, and precision are used for the test set validation. First, all the four ML models, such as LR, RF, XGB, and DT, are trained on 8 feature sets obtained by feature selection techniques. Feature selection techniques are applied to obtain different feature sets. Let F_i represent the i-th feature set. Eight feature sets denoted as $F_1, F_2, ..., F_8$, are obtained using different feature selection methods. PCA is applied to further reduce the dimension from 8 to 6.

Machine learning classifiers

The combination of advanced feature selection with ensemble methods forms the complementary framework for dealing with issues of dimensionality and accuracy. Techniques like Chi-square and Info Gain perform the dimensionality reduction on the data very systematically to remove all those features that do not positively impact the training algorithm. This refinement enables ensemble methods, such as XGBoost and RF ensembles, to run more effectively while leveraging their underlying architectures to model complex data relationships and reduce overfitting. As such, not only does the hybrid model improve the weighting of features central to CVD classification, but it also improves model generalization. While maintaining proper interpretability and improving predictive accuracy, it is highly appropriate for classifying CVD. The details of classifiers used are as follows:

Decision tree

A Decision Tree (DT) helps to understand a data relationship in a tree hierarchy structure [59]. A simple dataset and fewer features make it quick and practical, thus producing better predictions. The main reason for selecting the DT is that finding hidden trends and associations within the data set is very useful [60]. It is one way of representing an algorithm containing only a conditional control statement. The tree-based learning approach makes predictive models highly accurate, robust, and easily understood. It also needs less effort to prepare the data during the preprocessing phase [61]. Survey data often contain missing values, whereas building a DT is substantially less affected by missing values. The other main reason to use this algorithm is that it is possible to simplify the visualization of a complex DT system. The number of hyper-parameters to be modified is almost zero and supports multiclass classification problems. Algorithm 2 presents a DT for Heart Disease Classification, whereas Table 6 provides the symbol details. The algorithm describes building a DT model for classifying heart disease based on a given dataset D with features X and a target variable Y. The algorithm consists of two main procedures: *BuildDecisionTree* and *Predict*. In *BuildDecisionTree*(D, X, Y), the

Table 6 Symbols used in DT algorithm

Symbol	Description
D	Dataset
X	Set of features
Y	Target variable
T	DT model
A	Attribute/feature
v	Value of an attribute/feature
y	Class label
T_v	Subtree corresponding to $A = v$
x	Test instance
y_{pred}	Predicted class label

DT is recursively built by splitting the dataset based on different features until a stopping criterion is met. If all instances in D belong to the same class y, a leaf node is created with the class label y and returned. Moreover, suppose no more features are to split, or the stopping criteria are met (e.g., maximum tree depth, the minimum number of instances in a node). In that case, a leaf node is created with the majority class label in D and returned. The best feature A is selected using a splitting criterion (e.g., information gain, Gini impurity). A new DT node T is created with attribute A. For each possible value v of feature A, the dataset D is partitioned into subsets D_v based on $A = v$, and if D_v is empty, a leaf node is attached to T with the majority class label in D. Else, the subtree $BuildDecisionTree(D_v, X \setminus A, Y)$ is recursively attached to T and the DT T is returned. The Procedure, $Predict(T, x)$, predicts the class label for a given instance x using the trained DT T. It decides if T is a leaf node, the class label of T is returned, it determines the attribute A of T, and the value v of x for feature A is obtained. If v is not defined for A, the class label of the majority class in T is returned, $Predict(T_v, x)$ is recursively called, where T_v is the subtree of T corresponding to $A = v$. After that, in the training phase, An empty DT T is initialized and built by calling $BuildDecisionTree(D, X, Y)$. Whereas, in the testing phase, $Predict(T, x)$ is called to obtain the predicted class label y_{pred} and y_{pred} is assigned as the predicted class label for each test instance x.

Algorithm 2 Decision tree for heart disease classification

```
Input: Dataset D with features X and target Y
Output: Trained DT model T
procedure BuildTree(D, X, Y)
    if All instances in D belong to the same class y or stopping criteria met then
        return a leaf node with the majority class in D
    end if
    Select best feature A using a splitting criterion
    Create a node T with attribute A
    for each value v of A do
        Partition D into D_v where A = v
        Attach BuildTree(D_v, X \ {A}, Y) to T
    end for
    return T
end procedure
procedure Predict(T, x)
    if T is a leaf node then return class label of T
    end if
    Follow the branch of T based on x's value for A
    return Predict(T_v, x), where T_v is the corresponding subtree
end procedure
Training: T ← BuildTree(D, X, Y)
Testing: For each x ∈ test set, predict y_pred ← Predict(T, x)
```

Random forest

Combining Ensemble Learning techniques with the Begging algorithm, RF is used. It generates a set of DTs by selecting subsets of the training sets at random. It determines the ultimate class of the test object by aggregating the ballots from various DTs [38]. This aggregation of numerous DTs produces more precise outcomes by diminishing the variance and influence of noise. Moreover, this approach mitigates the issue of model overfitting. RF trains on a random subset of the training samples and attributes after constructing multiple DTs [62]. The primary justification for utilizing RF is its exceptional stability, which is significantly unaffected by the addition of new data points. It operates efficiently with categorical and continuous variables and handles absent values automatically. It is also exceptionally resistant to outliers. RFs are also capable of processing massive amounts of data containing thousands of variables [63]. Datasets are automatically balanced when a particular class occurs less frequently than other classes, rendering them appropriate for intricate tasks. Algorithm 3 represents the RM for heart disease classification, and Table 7 provides the details and descriptions of the symbols used. The algorithm builds multiple DTs from several randomly chosen subsets of the training data to produce an RF model. Due to the random selection of subsets, each DT in the forest captures a unique aspect of the data and offers a varied set of classifiers. Combining the predictions of different DTs, the RF approach seeks to decrease overfitting and enhance overall predictive performance. The resulting RF model can be applied to new, unforeseen instances to classify heart disease. The algorithm describes creating an RF model using a training set (S) and a set number of trees (N) to categorize heart diseases. It produces an ensemble of DTs by randomly choosing subsets from the training set S and using them to construct individual DTs. The DTs forest is initialized in an empty set F. A subset S' from the training set S with replacement is chosen randomly for each tree iteration, i. It enables variation in the training data for each tree because each instance in S' is chosen independently and may be selected more than once. A DT T_i is built with the subset S'. Finally, the forest F is built with the DT T_i.

Algorithm 3 RF for heart disease classification

```
Input: Training set S, number of trees N
Output: RF model
procedure RandomForestHeartDisease(S, N)
    Initialize an empty set F for the forest of DT
    for i = 1 to N do
        Randomly select a subset S' from S with replacement
        Create a DT T_i using S'
        Add T_i to the forest F
    end for
    return F ( RF model)
end procedure
```

Table 7 Symbols used in the RF algorithm for heart disease classification

Symbol	Description	Meaning
S	Training set	The original dataset containing samples of heart disease patients used for training the RF model
N	Number of trees	The desired number of DT to be included in the RF for heart disease classification
F	RF model	The ensemble of DT created by the RF algorithm for heart disease classification
S'	Subset of the training set	A randomly selected subset of the training set, with replacement, used for building an individual DT
T_i	DT	An individual DT within the RF for heart disease classification

XGBoost

The XGBoost algorithm is built on the principle of the gradient boosting framework to produce a prediction model as an ensemble of prediction models. It is just like other boosting methods and builds the model stage-wise. XGBoost is a more standardized form of gradient boosting; however, it is unique because it uses advanced regularization to control overfitting and improves its generalization capabilities [8]. One of the reasons for choosing XGboost is that it has built-in L1 and L2 regularization to prevent model overfitting. Two hyper-parameters: alpha and lambda, are used for L1 & L2 regularization. Also, it has the built-in capability to handle missing values. XGBoost is highly parallelizable/Scalable, quick in execution, and outperforms other algorithms [64]. Algorithm 4 describes how to build an XGBoost model using a training set ($\mathcal{D}$) and a predetermined (M) number of boosting iterations. The algorithm sets the model's initial condition to $F_0(x) = 0$. The pseudo-residuals r_{im} for every instance in the training set are then calculated after iterating through every boosting iteration. To create a weak learner $h_m(x)$, instances and their corresponding pseudo-residuals are trained on a weak learner (e.g., DT). The model is revised by adding the weak learner's scaled predictions to the original one, with the scaling factor η (i.e., learning rate). Finally, after all boosting iterations, the XGBoost model $F(x)$ is returned. The algorithm uses weak learners' additive properties to boost the model's predictive capability gradually. The description of all the symbols used in this algorithm is provided in Table 8.

Algorithm 4 XGBoost for heart disease classification

1: **Input**: Training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, number of boosting iterations M
2: **Output**: XGBoost model $F(x)$
3: Initialize model $F_0(x) = 0$
4: **for** $m = 1$ to M **do**
5: Compute pseudo-residuals: $r_{im} = y_i - F_{m-1}(x_i)$ for $i = 1$ to N
6: Fit a base learner (e.g., DT) to (x_i, r_{im}) to obtain a weak learner $h_m(x)$
7: Update the model: $F_m(x) = F_{m-1}(x) + \eta h_m(x)$, where η is the learning rate
8: **end for**
9: **Return** XGBoost model $F(x) = F_M(x)$

Table 8 Symbols used in XGBoost algorithm

Symbol	Description
$\mathcal{D}$	Training set
(x_i, y_i)	Training example with features x_i and label y_i
N	Number of training examples
M	Number of boosting iterations
$F(x)$	XGBoost model
$F_0(x)$	Initial model
r_{im}	Pseudo-residual for example i at iteration m
$h_m(x)$	Weak learner (e.g., DT) at iteration m
η	Learning rate

K-nearest neighbors

A supervised learning algorithm predicts by calculating the distance b/w training examples and new test instances. It is a lazy and non-parametric learning algorithm that makes no assumptions about the underlying data [65]. It just stores data, and when a new instance comes for testing, it starts working on it. This method works on the bases of majority votes. First, it calculates the distances b/w training examples and new test instances [62]. Based on nearest neighbors, which is the value of 'k,' examples with minimum distance are considered. If the value of 'K' is 3, it will consider the first three minimum values for the given class and assign the new instance to the majority class. The proposed model works better when the amount of data is less, and the number of features within the dataset is small. The dataset used in this project is small, and after feature selection, the dimension of the dataset will decrease further. KNN will work well with fewer features and data points. KNN, in our case, is presented in Algorithm 5. With the help of a training set ($\mathcal{D}$) and a set number of neighbors (K), this algorithm describes how to predict the class label of a given test instance (x_{tt}). The first step of the algorithm is to calculate the distances, using a distance metric, such as Euclidean distance, between each training instance x_i and the test instance x_{tt} in the dataset. For every i, these distances are kept in a list d_i. The training instances are then

arranged according to how far they are from the test instance in ascending order. The algorithm selects the top K nearest neighbors based on the sorted distances. Finally, the predicted class label $\hat{y}tt$ for the test instance $x\ tt$ is chosen by a majority vote among the class labels of the K nearest neighbors. As a result, the algorithm returns the predicted class label $\hat{y}_{tt}$. It is an easy yet efficient classification algorithm because it takes advantage of the idea that instances with similar feature values tend to belong to the same class. Table 9 describes symbols used in this algorithm.

Algorithm 5 KNN for heart disease classification

Input: Training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, test instance x_{tt}, number of neighbors K
Output: Predicted class label $\hat{y}_{tt}$
Compute distances: $d_i = distance(x_i, x_{tt})$ for $i = 1$ to N
Sort training examples in ascending order of distances
Select the top K nearest neighbors based on distances
Compute majority vote among the K nearest neighbors to determine the predicted class label $\hat{y}_{tt}$
Return $\hat{y}_{tt}$

Logistic regression

In its simplest form, LR is a predictive modelling technique that uses a logistic function to characterise a binary dependent variable; however, other advanced variations exist. Estimating the logistic model parameters is at the core of LR, a subset of regression analysis [8]. A dependent variable in a binary logistic model can take one of two potential values, such as passing or failing, represented by a variable called an indicator having two values: "0" and "1". The logarithm of the probabilities for the "1" value in the logistic model is the linear combination of the independent variables, which may be continuous or binary. The probability of the value "1" occurring could range from 0 to 1. With a threshold for each independent variable, increasing one of the independent components gradually raises the probability of the specified outcome at an exponential rate; thus, the logit serves as the unit of measurement for the log-odds scale. This approach involves generalizing the likelihood ratio of a binary dependent variable [66].

Table 9 Symbols used in KNN algorithm

Symbol	Description
$\mathcal{D}$	Training set
(x_i, y_i)	Training example with features x_i and label y_i
N	Number of training examples
x_{tt}	Test instance
K	Number of neighbors
d_i	Distance between x_i and x_{tt}
$\hat{y}tt$	Predicted class label for $x\ tt$

LR for Heart Disease Classification is described in Algorithm 6. It describes how to train an LR model with the help of a predetermined training set ($\mathcal{D}$), learning rate (α), and a predetermined number of iterations (T). The algorithm iterates through the training process while initializing the weights vector ($\mathbf{w}$) to zeros. The logits and predicted probabilities for each training instance are computed in each iteration, along with the gradients based on the true labels and predictions, and the weights are updated accordingly. After completing the required number of iterations, the algorithm returns the learned LR weights ($\mathbf{w}$). The relation between features and binary class labels is modeled using the logistic function in the standard classification algorithm known as LR. By computing the dot product of the trained weights and features and using the logistic function to obtain predicted probabilities, the trained weights can then be used to make predictions on new instances. The description of symbols is given in Table 10.

Algorithm 6 LR for heart disease classification

Input: Training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, learning rate α, number of iterations T
Output: LR weights $\mathbf{w}$
Initialize weights: $\mathbf{w} \leftarrow 0$
for $t = 1$ to T **do**
    Compute logits: $z_i = \mathbf{w}^T \mathbf{x}_i$ for $i = 1$ to N
    Compute predicted probabilities: $p_i = \frac{1}{1+e^{-z_i}}$ for $i = 1$ to N
    Compute gradients: $\nabla_{\mathbf{w}} = \frac{1}{N} \sum_{i=1}^{N} (\mathbf{x}_i^T (p_i - y_i))$
    Update weights: $\mathbf{w} \leftarrow \mathbf{w} - \alpha \nabla_{\mathbf{w}}$
end for
Return $\mathbf{w}$

Neural networks

An ANN is a simulation to reenact a neuron's network that makes up a human mind so that the computing unit will become capable enough to learn and understand things and, based on that understanding, replicate the thought process that would enable it to settle for a choice. NN utilizes multiple layers of Numerical computation to figure out the processed data. This layer collects various environmental inputs. The ANN network then attempts to derive

Table 10 Symbols used in LR algorithm

Symbol	Description
$\mathcal{D}$	Training set
$(\mathbf{x}_i, y_i)$	Training example with features $\mathbf{x}_i$ and label y_i
N	Number of training examples
$\mathbf{w}$	LR weights
α	Learning rate
T	Number of iterations
z_i	Logit for example $\mathbf{x}_i$
p_i	Predicted probability for example $\mathbf{x}_i$
$\nabla\mathbf{w}$	Gradients of the weights $\mathbf{w}$

insights from that data. It has a chain of layers that are all interconnected with one another [67]. Like a human brain, these connections have different strengths depending on how heavily they are weighted. NN has different layers: input, hidden, and output. At the start, we pass weights, and then a bias value is added to the total calculated values. An activation function (sigmoid, relu, leaky relu, tanhetc) is applied to add non-linearity to the NN.

Algorithm 7 explains how to train an NN model for heart disease classification. It requires an input-output pair training set ($\mathcal{D}$), the learning rate α, the number of hidden layers L, the number of neurons per hidden layer M, and the overall number of iterations T. The algorithm starts by initialising the weights ($\mathbf{W}^{(l)}$) and biases ($\mathbf{b}^{(l)}_{l=1^L}$). After that, a loop with the predetermined number of iterations is run. The algorithm computes the predicted output ($\hat{y}_i$) of the network for each input (x_i) by performing forward propagation on iterations of the training set. Using a suitable loss function, such as $\mathcal{L}(\hat{y}_i, y_i)$, it uses the predictions to calculate the loss (*Loss*), which quantifies the difference between the predicted output and the actual label (y_i). The algorithm then performs backpropagation, computing the gradients of the loss concerning the weights ($\nabla_{\mathbf{W}^{(l)}}$) and biases ($\nabla_{\mathbf{b}^{(l)}}$). These gradients show the magnitude and direction of changes that must be made to the network's parameters to reduce loss. Then, the weights and biases are updated using the gradients and learning rate. The network can then learn from the training data and modify its parameters to enhance its predictions by repeating this process for the designated number of iterations. Last but not least, the algorithm returns the learned biases and weights. Using forward propagation through the network layers, these learned parameters can be applied to predict new instances. Table 11 provides the symbols and their meanings used in this algorithm.

Algorithm 7 NN for heart disease classification

Input: Training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, learning rate α, number of hidden layers L, number of neurons per hidden layer M, number of iterations T
Output: NN weights and biases $\{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_{l=1}^{L}$
Initialize weights and biases: $\{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_{l=1}^{L}$
for $t = 1$ to T **do**
  for $i = 1$ to N **do**
    Perform forward propagation:
    for $l = 1$ to L **do**
      Compute activation: $\mathbf{a}^{(l)} = \sigma(\mathbf{W}^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)})$
    end for
    Compute predicted output: $\hat{y}_i = \mathbf{a}^{(L)}$
    Compute loss: $Loss = \mathcal{L}(\hat{y}_i, y_i)$
    Perform backpropagation:
    for $l = L$ to 1 **do**
      Compute gradients: $\nabla_{\mathbf{W}^{(l)}} = \frac{\partial Loss}{\partial \mathbf{W}^{(l)}}, \nabla_{\mathbf{b}^{(l)}} = \frac{\partial Loss}{\partial \mathbf{b}^{(l)}}$
      Update weights: $\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \alpha\nabla_{\mathbf{W}^{(l)}}, \mathbf{b}^{(l)} \leftarrow \mathbf{b}^{(l)} - \alpha\nabla_{\mathbf{b}^{(l)}}$
    end for
  end for
end for
Return $\{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}_{l=1}^{L}$

Table 11 Symbols used in NN algorithm

Symbol	Description
$\mathcal{D}$	Training set
$(\mathbf{x}_i, y_i)$	Training example with features $\mathbf{x}i$ and label y_i
N	Number of training examples
$\mathbf{W}^{(l)}, \mathbf{b}^{(l)} l = 1^L$	Weights and biases of the NN
α	Learning rate
L	Number of hidden layers
M	Number of neurons per hidden layer
T	Number of iterations
$\sigma(\cdot)$	Activation function
$\mathbf{a}^{(l)}$	Activation values of layer l
$\hat{y}i$	Predicted output for example $\mathbf{x}i$
$\mathcal{L}(\cdot, \cdot)$	Loss function
$\nabla\mathbf{W}^{(l)}, \nabla\mathbf{b}^{(l)}$	Gradients of weights and biases

Experimentation setup, results, and discussion

The experimentation was done in a controlled setting and on computational platforms in order to have standardized results. The experiments utilize datasets obtained from UCI, a repository for ML. The proposed system configuration includes an Intel (R) Core (TM) GPU @3.20 GHz processor, a Windows 10 operating system, and 8.0 GB of RAM. These experiments employ Python as the programming language and Jupyter Notebook as the integrated development environment. A highly interactive web-based programming environment is in question. All the experiments were carried out in Python 3.8 using the Jupyter Notebook with the help of libraries Scikit-learn (v0.24.2), XGBoost (v1.4.2), NumPy (v1.20.3), and Matplotlib (v3.3.4). In data preprocessing, the Min-Max scaling for continuous variables and one hot encoding for categorical features were applied. The data set was divided into 80% training and 20% test sets, and the sampling technique used was stratified to maintain an almost equal ratio of the classes.

For Neural Network models, training was conducted to 50 epochs with a batch size of 32 and an adaptive learning rate. XGBoost and RF models were optimized using grid search, with details summarized in Table 12. Particular performance metrics will be utilized to assess the performance of each model. This assessment will aid in the selection of the most effective classifiers for subsequent predictions. The efficacy of each model is assessed by means of a confusion matrix. It specifies how classifiers perform when the true values of a set are known. The confusion matrix provided a count value summarization of the number of accurate and inaccurate predictions, with these values further categorized by class. It aids in comprehending the errors committed by the classifier. Accuracy, recall, F-1

Table 12 Experimental setup for CVD classification

Aspect	Details
Hardware	CPU: Intel Core i7-9700K @ 3.60GHz
	GPU: NVIDIA GeForce RTX 2080 Ti
	RAM: 32 GB
	Operating System: Ubuntu 20.04
Software	Programming Language: Python 3.8
	Libraries: Scikit-learn (v0.24.2), XGBoost (v1.4.2), NumPy (v1.20.3), Pandas (v1.2.4), Matplotlib (v3.3.4)
Dataset Handling	Stratified sampling for 80:20 train-test split
	Scaling: Min-Max normalization
	Encoding: One-hot encoding for categorical variables
Hyperparameter Tuning	*XGBoost:*
	- `n_estimators`: [50, 100, 150, 200]
	- `learning_rate`: [0.01, 0.05, 0.1, 0.2]
	- `max_depth`: [3, 5, 7, 9]
	RF:
	- `n_estimators`: [50, 100, 200]
	- `max_depth`: [None, 10, 20, 30]
Training Parameters	Neural Networks: 50 epochs, batch size 32, adaptive learning rate
	Models: Trained on 80% data, tested on 20%
Evaluation Metrics	Accuracy, Precision, Recall, F1-score
	Confusion Matrices

score, and precision are some of the metrics utilized to assess the algorithms [68]. To balance diagnostic performance against clinical relevance, the evaluation metrics, including accuracy, precision, recall and F1 score, were considered. For instance, precision reduces false positives, preventing unnecessary interventions and minimizing the likelihood of missing high-risk patients; recall is essential to ensure no high-risk patients are ignored. This metric combines the above into a valued view of a model's performance. Accurate and reliable predictions of these metrics are aligned with clinical priorities, and accurate and reliable predictions are necessary to enable timely interventions and improved patient outcomes. The selected evaluation metrics are relevant to the clinical importance of CVD diagnosis. They reduce the risk of classifying individuals without CVD as high-risk patients and thus avoid unnecessary tests. Recall stresses correct recognition of the patients who require further medical attention. The F1-score represents both of them because, in high-risk medical applications, accuracy and reliability are equally significant.

Accuracy

A measure of the efficacy of a classifier lies in how accurately it makes classification. It is calculated as the proportion of validated observations relative to the total number of observations. Its range of values is from 0 to 1. A precise value of 1 indicates the highest or optimal level of accuracy, while a value of 0 indicates the lowest level of accuracy. The accuracy of a model can be mathematically computed as [69]:

$$A_{CC} = \frac{(TR_{ps} + TR_{ng})}{(TR_{ps} + TR_{ps} + FL_{ps} + FL_{ng})} \tag{10}$$

Where A_{CC} represents the accuracy, TR_{ps} represents the True Positive value in the label class, which is true, and the model correctly identifies it as true. TR_{ps} represents the True Negative value in the label class, which is false, and the model correctly identifies it as false. FL_{ps} represents the False Positive value, which is true; however, the model incorrectly classifies it as false, and FL_{ng} represents the False Negative value as false; however, the model classified it as true.

Precision

The ratio of the sum of positive predictive values to the number of true positive predictive values represents precision. In addition, it returns a value from zero to one. A precision of 1 indicates the highest predictive or optimal value, while a precision of 0 indicates the lowest predictive value. Mathematically precision can be calculated as [70]:

$$pr = \frac{TR_{ps}}{(TR_{ps} + FL_{ps})} \tag{11}$$

Recall

The technical term for the recall is "Sensitivity", representing the ratio of accurate positive value classifications to the sum of multiple positive samples constituting it. Recall values range from 0 to 1, with 1 indicating a high recall and 0 indicating a poor recall. Mathematically it can be computed as [70]:

$$rc = \frac{TR_{ps}}{(TR_{ps} + FL_{ng})} \tag{12}$$

F1- measure

The F1 metric is accurate and retrieves the weighted average, also known as the F1-score, which considers false negatives and false positives. By attributing greater significance to precision and less to recall, the beta value is determined to be 0.75. By assigning greater significance to recall and lesser significance to precision, the beta value is determined to be 2. We choose the beta value '1' to achieve a balance between recall and precision. However, F1 is usually more beneficial than accuracy, especially when the class distribution is uneven. When false negatives and false positives have comparable costs, accuracy performs best. Considering Precision and Recall together is preferable if the costs of false positives and false negatives vary. Its value is between 0 (worst) and 1 (best). It shows a small value if the precision or recall is small. It is determined using the harmonic mean of precision and recall.

$$F1 - Score = 2 \times \frac{(prxrc)}{(pr + rc)} \tag{13}$$

Classifier results on the subset of 8 features

Multiple ML models such as RF, DT, LR, and XGBoost have been applied to classify CVD. The following is a detailed discussion of each model's performance in terms of accuracy, recall, F1-measure, and precision:

Result on chi-square

Out of 13 input features, chi-square has identified the 8 most essential features. The feature subset extracted through chi-square, that is, age, chol, fbs, thalach, cp, slope, exang, and trestbps, are passed to models, and their performances are evaluated. Figure 2 show the results of each classifier. Among all the models, the results of XGB are the highest. XGBoost reported 99% accuracy, F1-score, precision, and 100% recall on the subset of 8 features.XGBoost performs well because it works like a chain of trees where the first tree classifies the data, and then the second tree tries to convert the incorrect classification of the first tree, thus making the model strong.

The main focus of this study is to assess the influence of feature selection on model performance for a structured tabular dataset with 13 features regarding CVD diagnosis. It shows that feature selection had a marginal impact on improving the performance of most ML models. For example, the accuracy of the XGBoost classifier rose to 98% from 85.25% without feature selection, as shown in Table 13. Likewise, RF improved its performance from 86.89% to 93% without and with feature selection, respectively. This means that feature selection efficiently removes redundant or less informative features that allow the model to focus on the most informative attributes.

On the other hand, LR had similar results: 85.25% accuracy without feature selection and 80% with it. That again suggests that when dealing with simpler linear models, the complete feature set may return results as good as optimized subsets because of their relative immunity against overfitting. However, this study further shows that the requirement of feature selection for small data sets may depend on context. From a purely predictive standpoint, modern ML models can process all 13

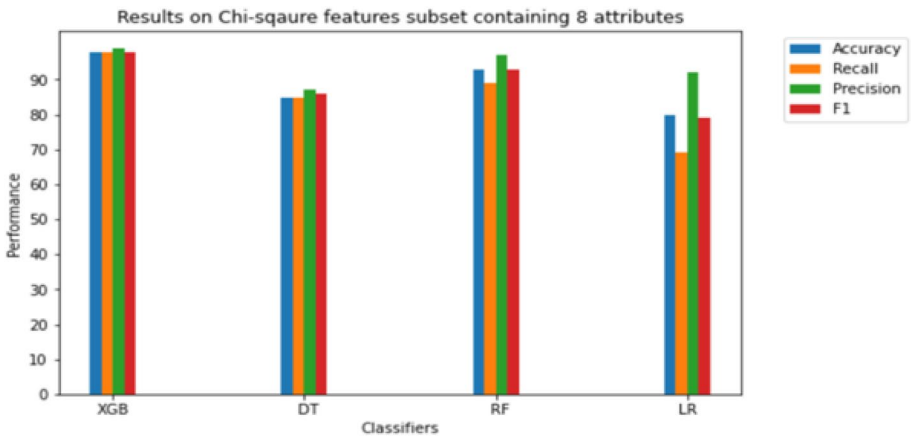


Fig. 2 ML models on Chi-square subset

Table 13 Comparison of Results With and Without Feature Selection (Chi-square)

	With feature selection	Without feature selection	With feature selection	Without feature selection
Classifier	**Accuracy**		**Recall**	
XGBoost	98%	85.25%	98%	84.38%
DT	85%	81.97%	85%	84.38%
RF	93%	86.89%	89%	87.5%
LR	80%	85.25%	69%	84.38%
Classifier	**Precision**		**F1-score**	
XGBoost	99%	87.10%	98%	85.71%
DT	87%	81.82%	86%	83.08%
RF	97%	87.5%	93%	87.5%
LR	92%	87.10%	79%	85.71%

features without a considerable loss of performance. However, feature selection is still vital for interpretability. Feature selection techniques reveal the most informative features, like cholesterol levels, exercise-induced angina, and maximum heart rate, which determine the factors influencing CVD diagnosis. Besides aligning the models with clinical intuition, it also ensures that the decision-making process of these methods is transparent and interpretable, which is significant in the ethical applications of AI.

Moreover, the results show that the models can be effective considering all features. Therefore, the proposed method would work fine without considering the numerous feature options. In conclusion, feature selection may be optional from a computational standpoint for small datasets. However, it certainly has an invaluable role in interpretability, reducing model complexity, and aligning the decisions taken by the model with clinical understanding.

Results on info gain

Features extracted through Info gain are age, slope, cp, chol, sex, thalach, exang, and old peak. Info gain is a filter method used to find and evaluate relationships between input and target features; no ML model is involved in such a method. It identifies significant features through correlation with the target class. Figure 3 and Table 14 show the results of each classifier. Among all the models, the results of XGB are the highest. XGBoost reported 99% accuracy, f1-score, precision, and 100% recall on the subset of 8 features. XGBoost performs well because it works like a chain of trees where the first tree classifies the data, and then the second tree tries to convert the incorrect classification of the first tree, thus making the model strong. From all the ML models assessed, XGBoost resulted in better performance metrics with accuracy, precision, recall, and F1-score. It was achieved due to its state-of-the-art gradient boosting framework, which works to classify samples by reducing the number of errors in classification on every successive round. Moreover, the L1 and L2 regularization techniques in XGBoost help to decrease overfitting, which often arises in DTs and LR. This difference in strategy allows XGBoost to capture complex patterns and interactions between features since trees are built sequentially, in contrast to the bagging strategy of RF. Further, its scalability and handling of missing data with an efficient hyperparameter tuning algorithm make it suitable for handling diverse and imbalanced datasets. These strengths make XGBoost a robust and reliable model for CVD classification.

Results on forward selection

Forward feature selection is an iterative strategy in which no feature is included at the start. The model adds a feature in each iteration and selects those features which optimize its performance. When there is no improvement

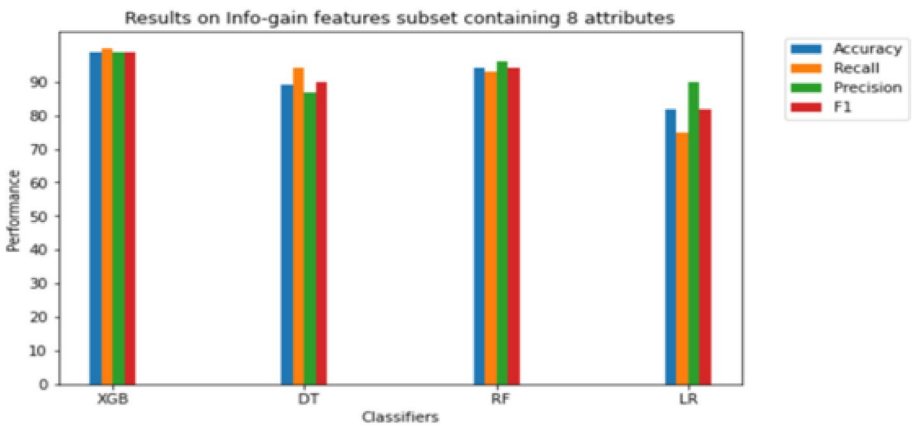


Fig. 3 ML models on Info-gain subset

Table 14 ML models on Info-gain subset

Classifier	Accuracy	Recall	Precision	F1-score
XGBoost	99%	100%	99%	99%
DT	89%	94%	87%	90%
RF	94%	93%	96%	94%
LR	82%	75%	90%	82%

Table 15 Results on forward selection feature subset

Classifier	Accuracy	Recall	Precision	F1-score
XGBoost	96%	95%	97%	96%
DT	86%	84%	89%	86%
RF	92%	93%	92%	93%
LR	86%	81%	91%	86%

in the model's performance, the model stops adding new features. With the help of forward feature selection, a set of 8 significant features have been obtained. The feature set selected by the forward method contains age, cp, sex, trestbps, restecg, old peak,thalach, and slope. It is evident in Table 15 and Fig. 4 that the results of XGB are the highest. RF scored closer to XGB; however, the overall performance of XGB is far better than other models. Accuracy, recall, precision, and f-score of ensemble techniques such as XGB and RF are high in contrast to other ML models because such models use multiple classifiers and are not dependent on just a single predictor.

Results on backward elimination

This method starts with all the features and eliminates the least significant feature. After each iteration, the model removes those features which are not necessary. It repeats this process until no improvement is observed when characteristics are removed. From the entire dataset, the backward elimination method has identified the 8 most essential features. The feature subset extracted through backward technique, i.e., age, cp, sex, exang, restecg, trestbps, old peak, and slope, are passed to ML models. Table 16 represents the results of ML models on the chi-square feature set. In this case, XGB outperforms other models by scoring 96% accuracy. Accuracy, recall, precision, and f1-score of ensemble techniques such as XGB and RF are high in contrast to other ML models because such models use multiple classifiers and are not dependent on just a single predictor. Moreover, ensemble methods work well on complex data. The chances of overfitting are less than other classifiers as it splits the data into multiple sets, and a tree is trained on each subset. From Fig. 5, it is clear that the performance of XGB and RF is better than DT and LR.

Classifier results on 6 features subset

Results on chi-square + PCA

Out of 13 input features, chi-square has identified the 8 most essential features. The feature subset was extracted through chi-square, i.e., age, chol, fbs, thalach, cp, slope, exang, and trestbps. This subset is further reduced into 6 by applying PCA, commonly used for dimension reduction purposes. Table 17 represents the results of ML models on the chi-square feature set. In this case, XGB outperforms other models by scoring 98%, 98%, 97%, and 100% in accuracy, F1-score, recall, and precision. Figure 6 shows below the performance of each ML classifier through different performance evaluation metrics. The results of XGB are the highest compared to the other ML models. Although DT reported a better recall score than RF and LR after XGB, the overall performance of RF is better than DT.



Fig. 4 ML models on forward selection subset

Table 16 Results on backward elimination feature subset

Classifier	Accuracy	Recall	Precision	F1-score
XGBoost	96%	94%	99%	96%
DT	87%	94%	84%	89%
RF	92%	87%	97%	92%
LR	84%	80%	88%	84%

Table 17 Results on PCA + Chi-square feature subset

Classifier	Accuracy	Recall	Precision	F1-score
XGBoost	98%	100%	97%	98%
DT	87%	99%	81%	89%
RF	93%	92%	94%	93%
LR	84%	78%	90%	84%

Results on info gain + PCA

Features extracted through Info gain are age, slope, cp, chol, sex, thalach, exang, and old peak. This subset is further reduced into 6 by applying PCA, commonly used for dimension reduction purposes. Table 18 and Fig. 7 represent the results obtained by ML models on the chi-square feature set. In this case, XGB outperforms other models by scoring 98%, 98%, 97%, and 100% in accuracy, F1-score, recall, and precision.

Results on forward selection + PCA

With the help of forward feature selection, a set of 8 significant features have been obtained. The feature set selected by the forward method contains age, cp, sex, trestbps, thalach, restecg, old peak, and slope. The feature subset is passed to PCA to further decrease the dimension before giving it into models. It is evident in Table 19 and Fig. 8 that the results of XGB are the highest. RF scored closer to XGB, but the overall performance of XGB is far better than other models.

Results on backward elimination + PCA

From the entire dataset, the backward elimination method has identified the 8 most essential features. The feature subset extracted through the backward technique is age, cp, sex, exang, restecg, trestbps, old peak, and slope. In order to reduce the feature subset, PCA is applied to the backward feature subset, which combines all 8 features into 6. Table 20 represents the results of ML models on the chi-square feature set. In this case, XGB outperforms other models by scoring 96% accuracy. In Fig. 9, it is clear that the performance of XGB and RF is better than DT and LR.

Figure 10 depicts the performance of the proposed model compared with the existing state-of-the-art. Other studies are compared to the proposed model regarding accuracy, precision, recall, and f-measure. This research focused on CVD detection, which is a death-taking disease. As shown in Fig. 10, the proposed IG_XGB performed better than the other state-of-the-art evaluation parameters mentioned above. We compared the proposed work for accuracy, and the results show that there is 1.02% more accuracy in IG_XGB compared to the work proposed in [8]. Whereas it is 10.754% more when compared to the work presented in [9]. Our feature set differs because we used an embedded technique (i.e., info gain) that resulted in better accuracy. In the case of precision comparison, IG_XGB and [8] performed 100% well, whereas there is a 10.753% more precision rate when compared to [9]. It is because they did not focus more on feature selection. They used a linear model for feature selection. In addition, we also compared the proposed IG_XGB with state-of-the-art for recall. The results show that the proposed IG_XGB is 2.04% better in recall

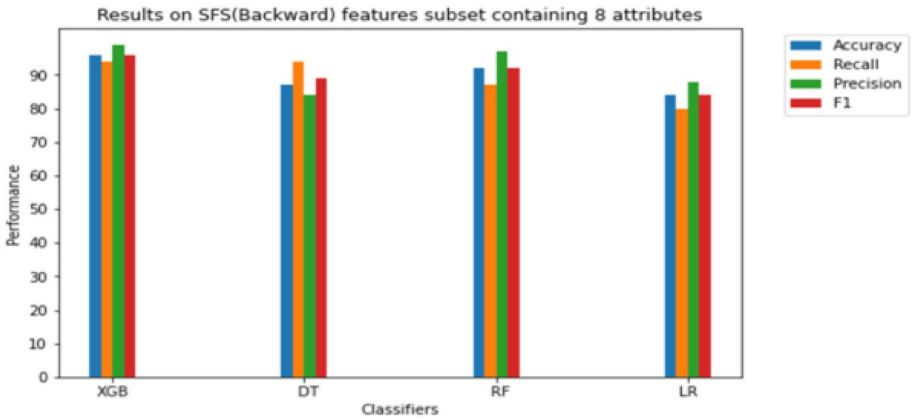


Fig. 5 ML models on backward elimination subset

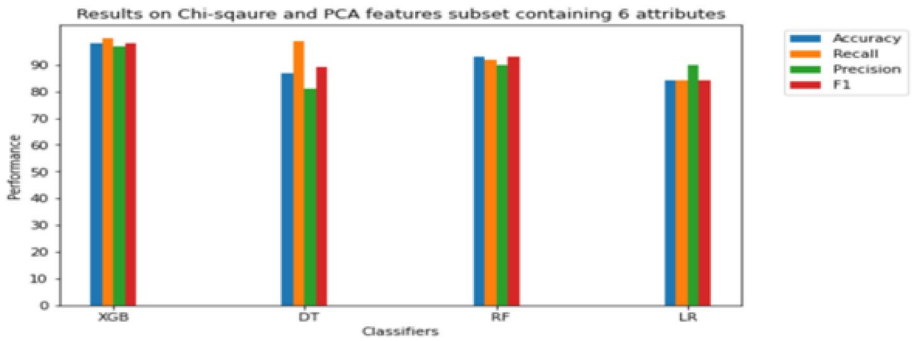


Fig. 6 ML models on PCA and Chi-square feature subset

Table 18 Results on PCA + Info gain feature subset

Classifier	Accuracy	Recall	Precision	F1-score
XGBoost	98%	100%	96%	98%
DT	86%	89%	85%	87%
RF	94%	89%	98%	94%
LR	83%	82%	85%	83%

than [8], and it is 6.25% more than [9]. Again, the reason is the excellent feature selection using info gain. Similarly, a comparison for the F1-score is also made to show the better performance of the proposed IG_XGB. The results show a 1.02% more F1-score value compared to [8], and it is 9.52% more when compared to [9]. Since the precision and recall for the proposed IG_XGB are better, the F1-score is also better.

To determine and preserve the combinations of features with maximum information, PCA was used on the original attributes to create the principal components, sorting features by their explained variances. Though yielding a minor amount of information loss, the trade-off was deemed worthy, considering the high accuracy and other performance metrics obtained when using the reduced feature set. Dimensionality reduction based on this latent space allows for better generalization and better interpretability and is specifically important in resource-constrained environments where computational efficiency is key. Future work will establish further findings and prove the effect of these trade-offs on clinical decision-making.

Confusion matrix

A detailed analysis of the classification performance of each classifier, based on the confusion matrix, is presented in Fig. 11. Results show that XGBoost had the highest sensitivity and specificity with 100 true positives, 98 true negatives, and only one false positive, and no false negatives. It shows its strong power to accurately recognize patients with CVD and reduce the risk of misclassification. XGBoost and RF also did well. They were able to classify 89 positive cases and 97 negative cases, with them having slightly higher false negatives than XGBoost. While DT achieved acceptable performance, it also produced the greatest number of false positives (i.e., 13) and false negatives (i.e., 15), suggesting its tendency to overfit on the dataset. However, with the ability to interpret and be efficient, LR had a recall rate of 69%, resulting in 45 false negatives. First, we provide results to illustrate that



Fig. 7 ML models on PCA and Info gain feature subset

Table 19 Results on PCA + Forward selection feature subset

Classifier	Accuracy	Recall	Precision	F1-score
XGBoost	98%	100%	97%	98%
DT	86%	82%	91%	86%
RF	94%	93%	95%	94%
LR	83%	88%	81%	84%

Table 20 Results on PCA + Backward elimination feature subset

Classifier	Accuracy	Recall	Precision	F1-score
XGBoost	97%	95%	100%	97%
DT	88%	95%	84%	89%
RF	93%	88%	97%	93%
LR	85%	91%	82%	86%

these ensemble methods are better at CVD classification, as demonstrated by generating a decent sensitivity and specificity balance.

When examining the incorrect classifications, some patterns of errors were observed. Errors were primarily localized to the cases of overlapping feature values between the classes like cholesterol, resting blood pressure, and maximum heart rate, to mention a few. These cases were generally in a grey area and were close to the decision boundary. Further, some overrides were noted when features were missing or had missing values like thalach (maximum heart rate). Although data preparation techniques such as stratified sampling and scaling reduced total misclassification, missing values occasionally encouraged a small level of noise, affecting the model's predictions. Therefore, future improvements will be made in feature engineering, where interaction terms and domain-specific variables will increase the differentiation of borderline cases. The model's decision-making capabilities can be improved by using imaging or clinical notes as complementary data. Furthermore, extending further into imputation strategies for handling missing data, including generative models, will likely eliminate noise and increase the study's robustness and generalisability. These measures seek to extend from the current models, specifically the ensemble methods, observed weaknesses to enhance diagnostic precision in applied contexts.

Potential, limitations, and future directions

In this study, several novel contributions to CVD classification research are introduced. It integrates state-of-the-art feature selection techniques, such as Chi-square, Info Gain, Forward Selection, and Backward elimination, and it applies PCA to refine the selected subsets of features to enhance the features selected with optimum parameter values for diagnostic and computational efficiency. Second, it concentrates on the consistency of AI techniques with ethical principles, such as transparency, fairness, and bias aversion, an essential yet often neglected aspect of medical AI applications. Third, we evaluate and rigorously vet several ML models and find that XGBoost is the optimal algorithm, striking a good balance between performance and generalizability. Together, these contribute to a bridge in research gaps for optimized feature engineering, ethical alignment, and robust classification in medical diagnostics.

The proposed classification model has great potential for integration into clinical settings. For example, it can help cardiologists analyze diagnostic data from patients, including demographics, medical histories, and test results, to identify patients at high risk of having CVD. It would provide consistent and accurate risk assessments, reducing the delay in care. Besides, its ability to handle missing data, generalization, and interpretable results makes it suitable for resource-limited settings where complete data may not always be available. For instance,

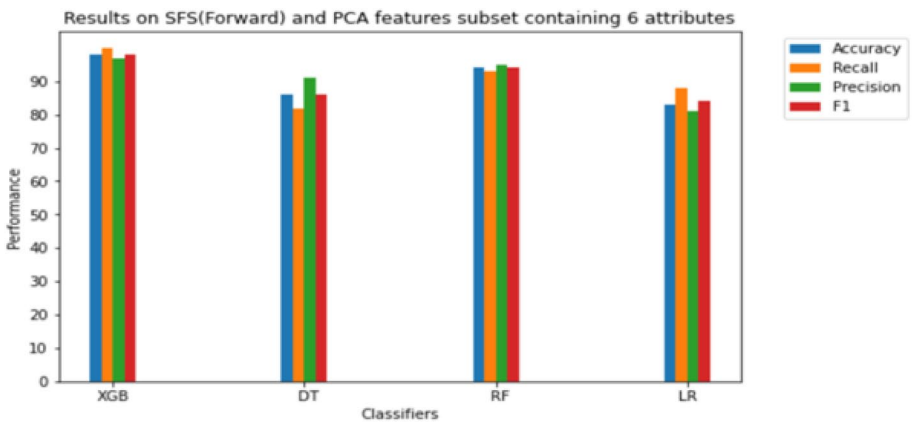


Fig. 8 ML models on PCA and Forward selection feature subset

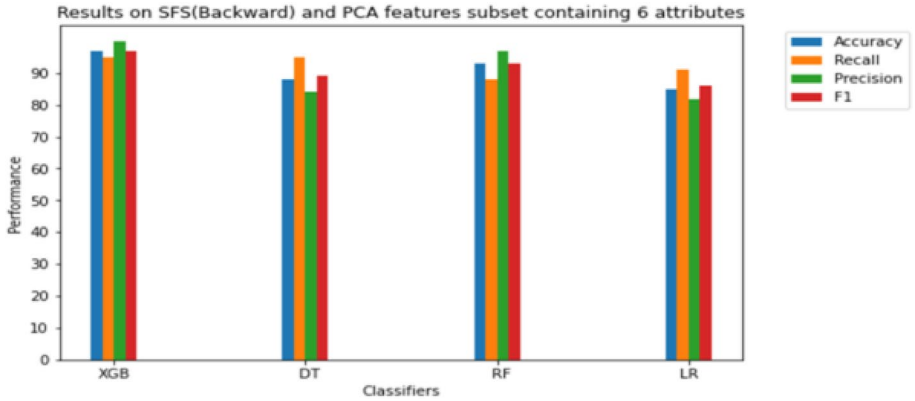


Fig. 9 ML models on PCA and Backward elimination feature subset

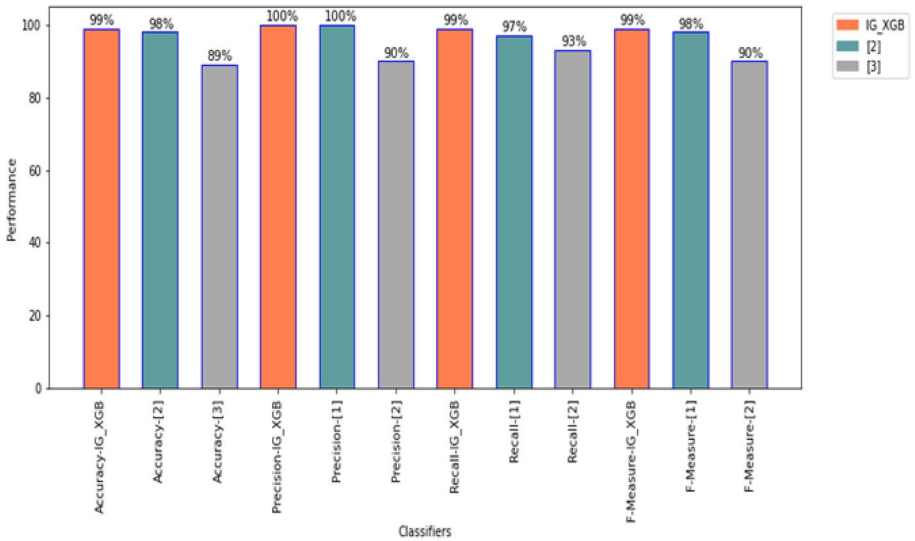


Fig. 10 Comparison of model performance with other studies

XGBoost had high accuracy, precision, and F1 score, but its complexity might make it overfit intricate patterns, especially on smaller and varied datasets. However, to address this risk, we optimized regularization parameters like λ and α using grid search and made sure that the model does indeed generalize well during training. The performance over various data splits was monitored through cross-validation, which reduced the chance of overfitting. The model is further verified with future validation on larger and more diverse datasets. It has great potential to reduce cardiologists' workloads in initial diagnostic assessments and, therefore, increase efficiency in healthcare settings. Its ability to handle missing data and continue to predict consistently is ideal for resource-limited areas, where it can expand access to quality diagnostics in underserved regions. Though XGBoost proved to be the highest-performing model, it has a complex runtime and is sensitive to hyperparameter tuning, which can be difficult with small or noisy datasets due to the risk of overfitting. Further investigation of other ensemble techniques or hybrid models, for example, stacking, might shed further light on model robustness.

The algorithmic complexities of the proposed framework's components justify the proposed framework's computational efficiency. XGBoost and RF are two such ensembles where training is logarithmic about the size of the dataset and tree depth ($O(T \cdot k \cdot \log(k) \cdot d)$) and linear in prediction time ($O(T \cdot d)$) making it feasible for real-time operations. LR, which is very simple to implement, has a training time of $O(k \cdot n)$, and the time for inference is $O(n)$. Operations of Chi-square and Info Gain work in $O(k \cdot n)$, which makes them scalable to

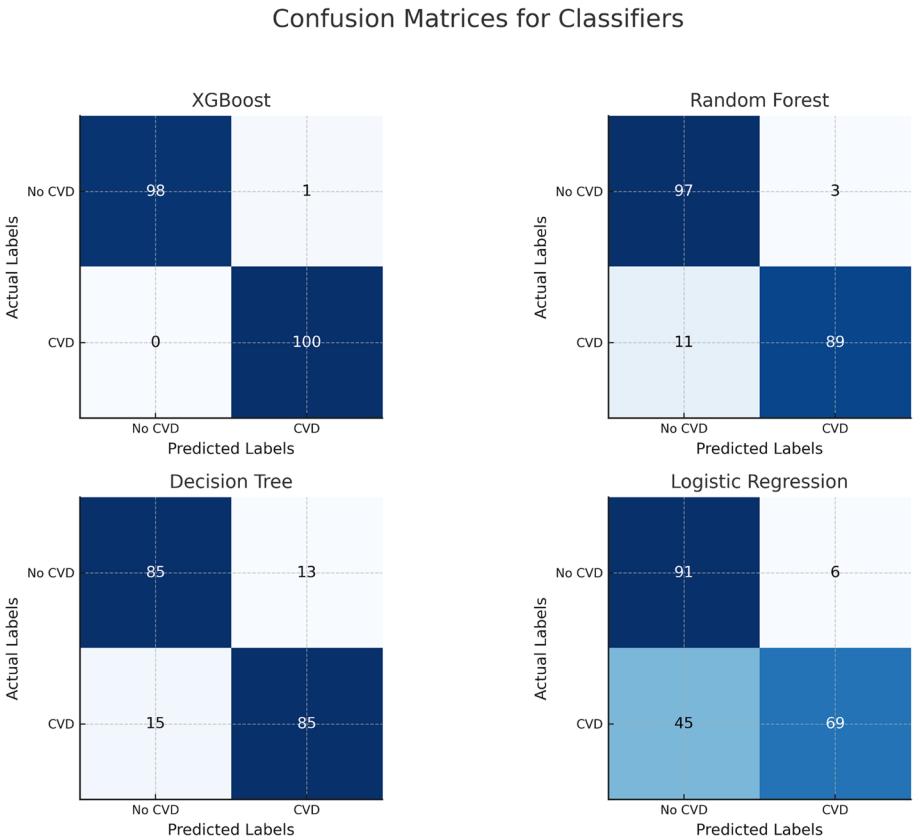


Fig. 11 Confusion Matrices

the number of features. At the same time, Forward and Backward Selection require $O(n^2 \cdot k)$ because of iterative computations. Training and testing PCA involves practical rather than theoretical trade-offs between variance preservation and computational cost; training requires $O(n^2 \cdot k + n^3)$ operations, while testing requires only $O(n \cdot r)$ operations where r is the number of selected principal components.

Besides risk prediction, the model can support personalized treatment planning by identifying unique risk factors for each patient. It aligns with the growing trend of precision medicine, where AI tools help tailor interventions based on individual patient profiles. In addition, the clinical workflows may reduce diagnostic bias and provide decision-making with an objective and data-driven perspective. These applications represent an area where AI systems make health care environments more efficient and improve patient outcomes; this is the doorway for future innovations. Several challenges exist in integrating the proposed model into clinical workflows: standardizing data across healthcare institutions, securing regulatory approvals, and training healthcare professionals to interpret what AI can tell them effectively. We ensure the model's explainability and alignment with ethical principles to create trust and adoption in clinical environments. Successful collaborations with clinicians and healthcare organizations will be dependent on this. In addition, some critical factors must be addressed to integrate the proposed framework into clinical workflows. The first requirement is to ensure it is compatible with existing hospital information systems (e.g., GPUs, cloud computing resources, or real-time processing). Second, healthcare providers' training programs are needed for effective adoption, emphasising reading model outputs and interfacing them with clinical decision-making.

Moreover, this study's methodologies are based on advanced feature selection and ML techniques to be robust and adaptable for different data. The datasets used for this study were acquired from the UCI repository and are not exhaustive of its global population variations. However, the proposed model shows the potential to generalize. It is robust due to advanced feature selection techniques and algorithms, which try to search and identify transferable features across datasets. However, the next step to bringing the model towards real utility and equitable diagnostic outcomes would be to validate the model's performance over more diverse datasets encompassing demographics more broadly. In addition,

enhanced preprocessing approaches, such as outlier analysis for managing large outliers and missing data management for actual clinical datasets should be included.

Similarly, though PCA successfully reduces feature sets, it can result in the loss of direct interpretation and crucial information from the original attributes. When considered with six attributes, the model maintained good performance. However, further research is recommended to validate the reduced feature set on different datasets to determine if this reduction affects the application of the model in various clinical settings. We also avoided class imbalance in the dataset by approximating class balance and also used stratified splitting to guarantee similar class distribution in the training and test sets. This approach helped to avoid potential sources of confounding, and the classifiers demonstrated high sensitivity and specificity with no need for further balancing methods. By maintaining the characteristics of the original data set, the methodology applied a fair approach in modelling and assessing the data for effective medical diagnosis. Future improvements to preprocessing should also include imputation strategies for missing values and domain-specific feature engineering to improve borderline cases and overlapping values.

While the current study indicates promising results, it also has some limitations. The data set was relatively small and applied to a specific demographic group; therefore, the generalizability of findings to diverse populations could be compromised. The datasets used in this study do not comprehensively represent global population diversity, which restricts the immediate impact of findings on diverse demographics. However, due to robust methodologies, the model shows strong potential for generalizability, and future work demands testing on more diverse datasets to generalize better. CVD characteristics vary considerably along age, gender, ethnicity, and geographic lines, which are not addressed in this work. Furthermore, because the feature selection techniques deployed are efficient, they may have failed to capture complex relationships arising with large or multimodal data sizes. In addition, real-world datasets in clinical settings have data missing or incomplete. Research in the future should incorporate those factors to make the model more robust.

Future studies should test the model's robustness and applicability for more diverse and heterogeneous data to overcome these inadequacies. It may include samples from different demographically varied settings and cardiovascular conditions. In addition, hybrid models of deep architectures will further improve accuracy and understandability, such as ensemble-based methods such as stacking and boosting, neural network hybrids like Neuro-Fuzzy systems and CNN-RNN architectures, and multimodal approaches that integrate structured data with medical imaging. Future work involves looking into longitudinal datasets concerning disease progression to predict the model's capabilities over time. It makes its utility much more in a dynamic healthcare environment. Therefore, it calls for advanced feature selection methods like transfer learning or domain adaptation for the complex data handling of the model and increases the generalizability of the model. An increase in scope towards a broader prediction of CVDs will further enhance the clinical utility of such models. In the future, external datasets from different geographic regions with different demographics, such as age, ethnicity, and gender, will be used for validation. Moreover, the model will be tested on longitudinal datasets to evaluate its predictive power over time and its suitability for global healthcare systems using datasets from resource-limited settings.

Researchers should incorporate multimodal data from medical imaging, genomic information, and clinical notes into diagnostic tools. Such AI systems can be developed to be explainable (e.g., SHAP or LIME) and align with ethical guidelines, emphasizing interpretability to facilitate trust and adoption in clinical settings [71]. Some examples include adding echo or electrocardiogram-type data and tabular data for better cardiovascular risk stratification. Also, federated learning techniques could help secure and privacy-preserving collaborations across institutions to create globally relevant AI models. These general directions aim to inspire future advancements that make AI-driven diagnostics more robust, scalable, and ethically aligned with healthcare practices. Moreover, designing AI systems with fairness and transparency will be indispensable for promoting equitable access to excellent diagnostics across the globe. Furthermore, future work addressing data privacy issues may explore secure approaches such as federated learning, where the institutions collaboratively train a model without patient sharing of sensitive data. In addition, advanced bias detection and mitigation techniques must be considered to limit disparities in diagnostic precision across diverse demographics.

Conclusion

Heart disease still ranks among the leading causes of death, requiring accurate and efficient diagnostic methods. This study evaluated multiple ML algorithms, focusing on ensemble techniques, such as RF and XGBoost, advanced feature selection methods, and dimensionality reduction using PCA. Among others, XGBoost outperformed, indicating the robustness of heart disease classification. Our study suggested that choosing the right balance between accuracy, interpretability, and computational efficiency is important when selecting ML models. While the findings are

promising, the study has some limitations that need to be addressed. Firstly, although balanced and diverse, the size of the dataset utilized is much smaller than large-scale datasets available in medical research. It restricts the scope of generalizability of results across broader populations. Secondly, the structured tabular data were studied, excluding other data modalities, such as medical imaging, genomics, and clinical notes. Finally, while emphasizing ethical AI principles, the study needs real-world clinical validations to be confirmed reliable and trustworthy in terms of its model. Hybrid models and state-of-the-art DL architectures can also be incorporated into clinical decision support systems, which offer clinicians real-time, interpretable information. In future, the proposed methodologies will be validated on datasets with diverse populations to ensure that the model is applied to diverse clinical settings globally. It will also include heterogeneous data modalities such as medical images, genomics, and clinical notes to further improve the model's transferability across different healthcare settings. The proposed methodologies will be tested in population datasets to confirm that the model has been implemented on a global scale in various clinical environments.

Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Network
AUC	Area Under the Curve
CFS	Correlation-Based Feature Selection
CVD	Cardiovascular Disease
DT	Decision Tree
DNN	Deep Neural Network
FN	False Negative
FP	False Positive
GA	Genetic Algorithm
GBT	Gradient-Boosted Tree
HD	Heart Disease
ILFS	Infinite Latent Feature Selection
KNN	K-Nearest Neighbors
LR	Logistic Regression
ML	Machine Learning
MLP	Multilayer Perceptron
NB	Naive Bayes
PCA	Principal Component Analysis
RF	Random Forest
RFE	Recursive Feature Elimination
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
XGB	Extreme Gradient Boosting

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Authors' contributions
G.A and F.H did conceptualization and modelling. N.T and M.S worked with medthodology and literature review. M.H and M.A fetched the results and did verification.

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Data availability
Data is available at https://archive.ics.uci.edu/dataset/45/heart+disease.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing Interests
The authors declare no competing interests.

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